Reg. No. | | | | | | | | | | [14CS-IS45]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE-2016

THEORY OF COMPUTATION

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Instructions: Answer any **FIVE** full questions choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART – A

1. a. Define the languages described by DFA and NFA. **02**

b. Let M be a DFA, under exactly what circumstance is $\in$ belong to L (M). **02**

c. Construct a DFA accepting each of the following language: **07**

i. Strings if a's and b's not having even numbers of a's and b's

ii. $L=\{a^n bba \mid n\geq 0\}$

iii. $\{w \mid w\in\{a,b\}' \; n_a(w) \bmod 3 > n_b(w) \bmod 2\}$

iv. Strings of a,b,c having aaa as substring.

d. Design a NFA that accepts the language (aa*(a+b)) and convert it to DFA. **03**

2. a. Consider the following $\in$-NEA **08**

δ	$\in$	a	b	c
→p	ϕ	P	q	r
q	P	q	r	ϕ
r	q	r	ϕ	p

i. Compute the $\in$-closure of each state

ii. Convert the automata to DFA

b. Prove that for every $\in$-NFA N there is a DFA D such that L(N)=L(D) **06**

3. a. Define regular expression. Write a regular expression for the following languages. **06**

i. $L= \{ ab^n w: n\geq 3, w=\{a,b\}^*\}$

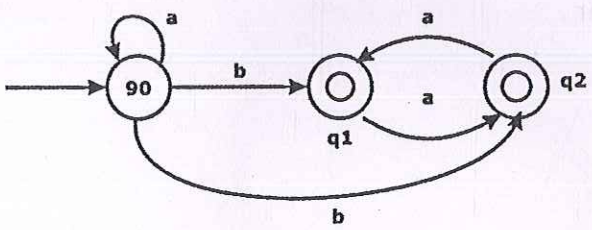
ii. All strings containing no more than three a's over Σ=a,b,c.

iii. $=\{vwv : v, w \in \{a,b\}^*, |v|=2\}$

iv. $L=\{w : w= \{a,b\}^* \; |w| \bmod 3=0\}$

b. Let r be the regular expression. There exists some nondeterministic finite acceptor M such that accepts L(r) =L (M). **04**

c. Find the regular expression for the language accepted by the following automata. **04**



4. a. Prove that regular languages are closed under intersection, reversal and complementation. **06**

b. Prove that the following languages are not regular. **04**

$L=\{a^p \mid p=$ Prime number$\}$

$L=\{WW^R \mid W \in \{a,b\}^*\}$

c. Minimize the following DFA. **04**

	0	1
→p	q	p
q	p	r
r	s	q
*s	s	p
t	s	u
u	v	t
v	u	v
w	v	s

PART-B

5. a. Define CFG, Sentence and language generated by grammar. **03**

b. Write CFG for the following languages. **07**

i. Strings of 0's and 1's for non-palindromes

ii. $L=\{o^m 1^n \mid m \neq n\}$

iii. $L=\{a^n b^m c^k : n+2m=k, n,m \geq 0\}$

iv. $L=\{a^i b^j c^k : i \geq j, k, i, j \geq 0\}$

c. Define ambiguous grammar. Prove that the following grammar is ambiguous. **04**

$s \rightarrow a|absb|aAb$

$A \rightarrow bs|aAAb$

6. a. Define a language generated by PDA using final state and empty stack. **02**

b. Construct a PDA accepting the language $\{a^m b^m c^n \mid m,n \geq 1\}$. Write instantaneous description for the string aabbccc. **08**

c. For the following grammar, construct a PDA: **04**

$S \rightarrow aABB \mid aAA$

$A \rightarrow aBB \mid a$

$B \rightarrow bBB \mid A$

$C \rightarrow a$

7. a. Consider the grammar **08**

$S \rightarrow ABC \mid BaB$

$A \rightarrow aA \mid BaC \mid aaa$

$B \rightarrow bBb \mid a \mid D$

$C \rightarrow CA \mid AC$

$D \rightarrow \epsilon | d$

i. Eliminate ϵ-Productions

ii. Eliminate unit productions from the resulting grammar

iii. Eliminates the useless productions from the resulting grammar

b. Define CNF. Convert the following grammar to CNF. **03**

$S \rightarrow ABa$, $A \rightarrow aab$, $B \rightarrow Ac$

c. Prove that context free languages are closed under Union, Concatenation and Star closure. **03**

8. a. Explain with neat diagram the working of a Turing machine model. **03**

b. Design a turing machine to compute the following function. **08**

$F(m,n)=m-n \ \{m \geq n\}$

$=0$ otherwise.

c. When is a problem said to be decidable? Give an example for undecidable problem. **03**